STPM Physics · Semester 1 · Chapter 2

Kinematics

2.1 Linear Motion

Definitions & laws

- **Distance** is the total length of the path travelled (scalar); **displacement** is the shortest straight-line distance from the initial to the final position in a stated direction (vector).
- **Speed** is the rate of change of distance; **velocity** is the rate of change of displacement (vector).
- **Instantaneous velocity** is the velocity at one instant, $v = \frac{ds}{dt}$, the gradient of the displacement–time graph at that instant.
- **Acceleration** is the rate of change of velocity, $a = \frac{dv}{dt}$ (vector).
- **Free fall** is motion under gravity alone, with constant downward acceleration $g = 9.81\ m\ s^{-2}$ near the Earth's surface (air resistance neglected), independent of mass.
- A v–t graph: gradient = acceleration; area under graph = displacement (area below the t-axis is negative). An s–t graph: gradient = velocity. An a–t graph: area = change in velocity.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Velocity–time equation	$v = u + at$	u = initial velocity; v = final velocity ($m\ s^{-1}$); a = constant acceleration ($m\ s^{-2}$); t = time (s); constant acceleration in a straight line only
Displacement equation	$s = ut + \frac{1}{2}at^2$	s = displacement (m); u = initial velocity ($m\ s^{-1}$); a = constant acceleration ($m\ s^{-2}$); t = time (s)
Velocity–displacement equation	$v^2 = u^2 + 2as$	u = initial velocity; v = final velocity ($m\ s^{-1}$); a = constant acceleration ($m\ s^{-2}$); s = displacement (m)
Displacement from mean velocity	$s = \frac{1}{2}(u + v)t$	s = displacement (m); u = initial velocity; v = final velocity ($m\ s^{-1}$); t = time (s)
Displacement from final velocity	$s = vt - \frac{1}{2}at^2$	s = displacement (m); v = final velocity ($m\ s^{-1}$); a = constant acceleration ($m\ s^{-2}$); t = time (s); valid for constant acceleration in a straight line only
Average velocity	$\frac{\Delta s}{\Delta t}$	Δs = change in displacement (m); Δt = time interval (s)
Average speed	total distance/total time	distance in m; time in s; speed in $m\ s^{-1}$

Quantity / law	Formula	Symbols, units & conditions
Vertical motion	$a = -g = -9.81\ \text{m s}^{-2}$	Take one direction positive (e.g. upward); g = gravitational acceleration; at the highest point v = 0

How to answer (MPM style)

- Sketch or state the sign convention first (positive direction), then list s, u, v, a, t with signs and pick the equation that omits the unknown.
- Write: formula → substitution with units → answer with unit, direction (sign) and 3 s.f.
- For graph work, state that the displacement is the area under the v–t graph and use the shapes (triangle ½bh, trapezium ½(a + b)h); for a curved s–t graph, draw a tangent for the instantaneous velocity.
- Sketch graphs with labelled axes and units, correct shape (straight line for constant slope, parabola for s–t when a is constant, horizontal line for a–t), intercepts and turning points.

Section A traps

Trap: Distance and displacement confused: on a v–t graph with a part below the axis, distance uses the sum of absolute areas, displacement uses the net area.

Trap: Using the equations of motion ($v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$) when acceleration is not constant (curved v–t graph or a changing a–t graph).

Trap: Sign of g: for a ball thrown upward with upward positive, $a = -g$; using $+g$ gives wrong time and height. Negative acceleration does not always mean slowing down.

Trap: At the top of the motion $v = 0$ but $a = g$, not zero; a body dropped from rest falls $s = \frac{1}{2}gt^2$, distance proportional to t^2, not t.

Tricks

Trick: Equal time intervals from rest give distances in the ratio 1 : 3 : 5 : 7 for constant acceleration.

Trick: Time up = time down for return to the same level (no air resistance); the speed on return equals the launch speed.

Trick: Sketch the v–t graph first: a straight line means constant acceleration, so the equations of motion for uniform acceleration apply; areas then give distances quickly.

2.2 Projectiles

Definitions & laws

- **Projectile** is an object projected into the air that moves under the action of gravity only (air resistance neglected).

- The horizontal and vertical motions are **independent**: horizontally constant velocity ($a_x = 0$); vertically constant acceleration g downward.
- The path (trajectory) of a projectile is a **parabola**.
- At the highest point the vertical component of velocity is zero, the horizontal component is unchanged, and acceleration is g downward.
- The speed at the same height on the way up and down is the same; the time of ascent equals the time of descent when landing at the launch level.
- A projectile fired horizontally has zero initial vertical velocity; its time of fall depends only on the height h, not on the horizontal speed.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Launch velocity components	$u_x = u\cos\theta;\ u_y = u\sin\theta$	u = launch speed (m s^{-1}); θ = angle above the horizontal; u_x, u_y = horizontal and vertical components
Velocity and position at time t	$v_x = u\cos\theta;\ v_y = u\sin\theta - gt;\ x = (u\cos\theta)t;\ y = (u\sin\theta)t - \frac{1}{2}gt^2$	v_x, v_y = velocity components (m s^{-1}); x, y = horizontal and vertical displacements (m); g = gravitational acceleration (m s^{-2}); t = time (s)
Vertical velocity–displacement	$v_y^2 = u^2\sin^2\theta - 2gy$	v_y = vertical velocity (m s^{-1}); y = vertical displacement (m); u = launch speed (m s^{-1}); g = gravitational acceleration (m s^{-2})
Speed	$v = \sqrt{v_x^2 + v_y^2}$	v = speed (m s^{-1}); v_x, v_y = velocity components (m s^{-1})
Angle below horizontal	$\varphi = \tan^{-1}\frac{\lvert v_y\rvert}{v_x}$	φ = angle of velocity below the horizontal; v_x, v_y = velocity components (m s^{-1})
Time to maximum height	$t = u\,\frac{\sin\theta}{g}$	t = time (s); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})
Time of flight	$T = 2u\,\frac{\sin\theta}{g}$	T = time of flight (s); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2}); landing at the same level
Maximum height	$H = \frac{u^2\sin^2\theta}{2g}$	H = maximum height (m); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})
Range	$R = \frac{u^2\sin 2\theta}{g}$	R = horizontal range (m); u = launch speed (m s^{-1}); θ = launch angle; R is maximum at θ = 45°, and θ and (90° − θ) give the same range
Trajectory	$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$	x, y = horizontal and vertical displacements (m); u = launch speed (m s^{-1}); θ = launch angle; g = gravitational acceleration (m s^{-2})

Quantity / law	Formula	Symbols, units & conditions
Horizontal launch from height h	$t = \sqrt{\frac{2h}{g}}$; $x = ut$	t = time of fall (s); h = height (m); x = range (m); u = horizontal launch speed (m s^{-1}); g = 9.81 m s^{-2}

How to answer (MPM style)

- Resolve the initial velocity first, then treat horizontal and vertical motions separately using the same time t as the link.
- State the sign convention for the vertical direction and use a = −g (or +g if downward is positive) consistently.
- Give the final velocity as magnitude and direction (angle below/above the horizontal); give distances to 3 s.f. with unit.
- For "explain" answers use: horizontal velocity is constant because no horizontal force acts; vertical acceleration is g because only weight acts.

Section A traps

Trap: Taking the horizontal velocity as zero at the top, or the acceleration as zero at the top; only v_y = 0 there.

Trap: Using $R = \frac{u^2 \sin 2\theta}{g}$ for a launch from a height or landing on a different level; the formula is for the same level only.

Trap: Using u instead of u cos θ in the horizontal distance, or u sin θ for the horizontal component when θ is given from the vertical.

Trap: With air resistance the range, maximum height and horizontal speed decrease, and the path is no longer symmetrical (steeper on the way down).

Tricks

Trick: Two objects, one dropped and one fired horizontally from the same height, hit the ground at the same time.

Trick: Check by limiting angles: θ = 90° gives R = 0; θ = 0° launched from a height gives time $\sqrt{\frac{2h}{g}}$.

Trick: At equal heights speeds are equal (energy conservation), so v can be found without solving for t.